

Summary: Proximal Origin — A Tragicomedy of our Times

This story is in 3 parts: [Summary¹](#), [Part 1²](#), [Part 2³](#)

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Tragic and comic masks of the Greek theatre — mosaic of Hadrian's villa

'And, unfortunately, you guys have to figure that out. I don't have to figure that out, but you guys have to figure it out.'

Introduction:

The correspondence entitled “The proximal origin of SARS-CoV-2”, published in the journal *Nature Medicine* on 17 March 2020, had a fundamental impact on the public and scientific conversations regarding the COVID-19 pandemic. As of September 2024, that correspondence is ranked #4 out of 26.5 mln research outputs tracked by Altmetric⁴, and #1 of around all 10,000 *Nature Medicine* outputs.

By asserting that “Our analyses clearly show that SARS-CoV-2 is not a laboratory construct or a purposefully manipulated virus,” that short correspondence played a major role in promoting a natural origin of the pandemic, after being relentlessly exhibited as a scientific denunciation of its alternative, the research-related accident hypothesis. Over the last three years, documents released through Freedom of Information (FOI) Act quer-

ies have steered an equally impressive amount of controversy on the genesis of that publication. Different versions and interpretations of what drove its writing and the evolution of its content have flourished — all of them equally unsatisfying.

The research that follows tells for the first time the true story of the genesis of Proximal Origin (P.O.), based on overlooked documentary evidence and a thorough examination of the context. For the first time also, the exact chronology of the FOI'd emails has been resolved, delivering additional insights.

Key questions are addressed in this research, such as:

- Who was really behind the investigative initiative of Jeremy Farrar, of the Wellcome Trust, at the end of January 2020.
- Why the tone and conclusions of the Proximal Origin draft changed so much within a week of the confidential investigative call that took place on 1 Feb, organised by Farrar at the suggestion of Anthony Fauci (head of NIAID).
- Why the agreed plan to have the WHO onboard the origin question was dropped around the 9 Feb, and never heard of again, despite having the support of the US government.
- Why Farrar organised both the publication of Peter Daszak's Statement of Support and the release of Proximal Origin at a crucial time.
- What the position and priorities of the US administration regarding the WHO, and getting access to data and isolates, were.
- What the role of Fauci in all of this was, and what he did in support of Farrar but also for his own sake.
- Who sent an anonymous letter to Jon Cohen at Science denouncing the Proximal Origin co-authors as impostors, in July 2020, and why.

As per the title, this is a true tragicomedy of our times, when international institutions are effectively being reshaped, with an assortment of well-meaning people, unreported blunders and impossible odds, that resulted in health diplomacy imperatives high-jacking a scientific enquiry and ultimately undermining science, public support for it, and the credibility of health authorities, from scientific journals to national and international organisations. That's the tragedy.

A few individuals, operating from the protection of their anonymity for their egotistic interests, also played an essential toxic role in the story of Proximal Origin. Last, and not to be underestimated, this research highlights how poor timings and excessive posturing in the US, including by the media, actually contributed to making the WHO work more difficult. Together, these form the comedy.

Let it be clear that the purpose of this research is not to hurt the WHO, but to show that, when not setting a clear and defensible course of action, and when confusing health diplomacy with science, things can quickly go wrong. We should leave diplomacy to the diplomats; many well-meaning characters, such as Farrar, and even initially Kristian Andersen, would have very likely been more comfortable, and ultimately more successful, had they stuck purely to science.

Last, when so many people have behaved poorly in this story, I wish to point out that Dr. Tedros (the Director General of the WHO), Ralf Baric (quoted above), and Ian Lipkin have behaved better than most, within the constraints they faced.

Warnings:

- The full article⁵ (in two parts) is rather long by necessity, so that reading it is better attempted after going through this summary of the key findings.
- Reading the full article (including the footnotes) is still required to fully understand the many twists and surprises in the story of Proximal Origin, as this summary leaves aside many important insights.

Podcast renderings:

A 6-minute Briefing and a full version of the Summary (31 minutes) are available in podcast format:

- **The full summary (31 minutes):**
- **A 6-minute briefing:**



Summary of key Findings:

A difficult start for the WHO (Jan)

In the second week of January 2020, the WHO found itself in a difficult situation. The Wuhan outbreak of a mysterious respiratory disease looked more and more like SARS-1, not just because it likely involved a coronavirus, but also because China was again starting on a rather uncooperative footing when it came to sharing data and relevant information.

Things came to a head when China withheld the publication of the sequence for weeks. A concerted effort between Jeremy Farrar (Wellcome Trust), Maria van Kerkhove (WHO), Eddie Holmes and his colleague Zhang Yong-Zhen (who had access to the sequence) eventually resulted in that sequence being published unilaterally, on 11 Jan, after China missed an ultimatum.

Farrar would soon intervene again, this time to force the disclosure of likely human-to-human transmissions, which was being vehemently denied by China against a surreal background of no new cases having been reported since early January. After receiving a tip, Farrar made sure that the WHO was alerted to these suspicions, looping in van Kerkhove again. The next day (20 Jan), China conceded human-to-human transmission.

After these two tense episodes, the outlook momentarily brightened up for the WHO, when it was able to send a first mission to Wuhan on 20–21 Jan. Immediately after, and potentially to show its goodwill, the WHO decided not to declare a Public Health Emergency of International Concern (22–23 Jan), even as Wuhan was put into a lockdown on 22 Jan.

The respite was short. Reported cases started exploding during the week of 23–28 Jan, spreading within China and to other countries. With all eyes now on locked-down Wuhan, the already controversial WHO decision not to declare a PHEIC on 22 Jan started looking rather bad to the international community. Dr Tedros tried to address the mounting challenges by flying to Beijing and meeting Xi Jin-ping (28–29 Jan), where he extracted some vague promise of cooperation and access to China for the WHO. Back in Geneva, a PHEIC was declared on 30 Jan.

So, after a fairly bad start that looked very much like a repeat of SARS-1, in the last days of January, the WHO could hope to be more in control of the situation, with the principle of essential access to China secured but still to be formalised. All without having had to take a proactive hard stance, but with a fair amount of fortuitous reactive work which had involved Farrar and Holmes, and all in the background.

The Farrar-MI5 side-check (last week of Jan)

Following the release of its sequence on 11 Jan, some concerns about a potential non-natural origin of the SARS-CoV-2 virus were raised internally in the US in the second or third week of January. On 25 or 26 Jan, Baric attended a classified BSEG (Biological Sciences Experts Group) meeting, organized by the National Counterproliferation Center (NCPC) within the Office of the Director of National Intelligence (ODNI). During that meeting, Baric laid out his three valid origin hypotheses for the outbreak: natural origin, laboratory escape and genetic engineering.

These non-public concerns soon reached the UK, as the main US partners of the US in the five-eyes, and eventually Jeremy Farrar of the Wellcome Trust. On 27 Jan, Farrar asked his trusted friend, Holmes, to quietly investigate some potential signs of a laboratory origin in the recently released sequence. This was meant as a confidential side-check, likely spurred and definitely supported by MI5.

Holmes, being an expert in coronaviruses and having been partly responsible for the release of the sequence, was soon contacted by Andersen, who had his own pointed doubts about the sequence. Andersen's main concerns were strengthened by a history of collaboration between the WIV in Wuhan, Baric in the US and EcoHealth Alliance under Daszak.

Fauci and the Farrar call (31 Jan-1 Feb)

For good process, Francis Collins (head of NIH) and Fauci (head of NIAID) were informed by Farrar. At the suggestion of Fauci, a conference call was then organised to review the findings of Farrar's core team, centred around Holmes and Andersen, in the presence of some additional experts (also at the suggestion of Fauci). These additional experts included GoF (Gain-of-Function) practitioners such as Ron Fouchier of the Netherlands and Christian Drosten of Germany, but excluded Baric and Daszak due to their conflicts of interest.

During the Farrar call on 1st Feb, the concerns of the core team of Andersen, Holmes, Andrew Rambaut and Robert Garry were met with derision and anger by the GoF practitioners, who successfully pushed back against the idea of deliberate engineering of the virus.

Following the call, Farrar's core team (but not the experts added for the purpose of the call at the demand of Fauci) started drafting a confidential report, the idea being to use it to offload the question of the origin to the WHO. Despite the deliberate engineering option being taken off the table during the call, the possibility of a lab origin was still present under the passaging hypothesis (a laboratory form of targeted and accelerated 'natural' selection), which had not been explored much on that occasion.

The Tedros Proposal (2 Feb)

The next stage was to contact the WHO to ask them to onboard the origin question, and to let the Chinese government know about it, in the best possible way, to defuse any overreaction.

Farrar first talked to Mike Ryan and Bernhard Schwartländer, Tedros' right arm, on 2 Feb, to brief them on the off-load proposal, the so-called 'Tedros proposal', and then got back to Collins. On the same day, Garrett Grigsby, HHS Director of Global Affairs, forwarded the Tedros proposal to Stewart Simonson, the Assistant Director-General of the World Health Organization responsible for WHO-US Liaison Office, who took over the contact with Schwartländer for the US government side. On 2 Feb also, Farrar contacted a former minister of health from the pre-Xi Jin-ping era, Chen Zhu, to let him know of the Tedros proposal. The relation with Chen Zhu went back to the time Schwartländer was working in Beijing.

So, within one day of the Farrar call during which the plan had been first laid out, Farrar and the US administration were trying to empower the WHO to take over the origin question.

Track II efforts, NASEM letter and footwork by Daszak (3-6 Feb)

At the time, the US administration was pushing hard to get strongly needed true epidemiological data and isolates. Towards this goal, the US administration implemented a combination of Track II diplomacy (US scientists to Chinese scientists, under the aegis of their respective national academies) and of careful support to the WHO. In parallel, at the

inter-governmental diplomacy level (Track I), Trump kept to his strategy of praising Xi Jin-ping and China (until the end of March), but very much with an eye on securing some trade concessions, and with little interest in the developing health crisis.

That Track II approach resulted in the Office of Science and Technology of Policy (OSTP, White House), ostensibly asking the National Academy of Sciences, Engineering and Medicine (NASEM) to come back with recommendations as to what type of data would be required to better understand the origin and likely evolution of the virus. Daszak and Andersen were part of the NASEM committee of experts involved.

Most likely, Daszak faced opposition from Andersen during a NASEM meeting on 3 Feb, held in the presence of some FBI and ODNI experts. After Daszak tried to get some strong natural-origin wording included in the NASEM answer, his wording did not make it to the final version sent to the OSTP on 6 Feb. That answer to OSTP included about three lines about desirable data, plus yet another reiteration of the need for China to share data, and had nothing more to say, having done its preaching job.

Beyond working on the NASEM letter with Andersen, the pro-lab accident stance that Farrar's core team had displayed on the Feb 1 call would have easily got back to Daszak via Fouchier or his colleague Marion Koopmans (also present on that call). These two shared Daszak's concerns about not restricting the specific field of rather risky virological research in which they were involved.

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Aware of that stance, and based on his NASEM letter experience, Daszak would have got particularly concerned that the report being drafted by the Farrar core-team would not fully close out the possibility of a lab origin; a potential devastating blow for his current grants, but also for his Global Virome Project (GVP) hopes. Daszak thus designed a pre-emptive movement with his 'Statement of Support', which he started drafting around 6 Feb 2020, while being careful to hide his leading role in it.

Fauci's mishap and death of the Tedros Proposal (6 Feb)

On 5 Feb, the WHO gave its green light to the Tedros proposal, and was willing to discuss how to onboard the origin question during the WHO Blueprint meeting to be held on 11–12 Feb. The next day (6 Feb), Farrar started feeding names for a WHO committee that should pick up the question.

On 6 Feb too, ABC published an article with some poor reporting and a clumsy quote by Fauci, that was very counter-productive to the Tedros proposal.

That article misrepresented the OSTP letter to NASEM as being a request by the US government to US scientists, asking them to investigate the origins of SARS-CoV-2. In fact, it was really nothing more than some Track II messaging to China, disguised as a request for a specification of the kind of data that would be useful, which ended up producing about three lines of data recommendations appended to a demand to share data.

Even worse, the article quoted Fauci as saying:

‘There’s always that concern,’ Dr. Anthony Fauci said [about the possibility of an engineered virus]. “And one of the things that people are doing right now is very carefully looking at sequences to see if there’s even any possibility, much less likelihood, that that’s going on. And you could ultimately determine that. So people are looking at it’.

So, for all purposes, the OSTP request was described as an investigation, while Fauci was confirming that investigation and was confident that US scientists could get to the bottom of the origin question with the right data. This sent the entirely wrong signal to the Chinese government, just when the actual plan was to offload the origin question to the WHO, to do so in a non-threatening way to China, and precisely at the time the WHO was trying to get a mission on the ground in China.

While the mishap was quite incredible for someone like Fauci who should have known better, we should not exaggerate its consequences: in the end, China would never have allowed for an investigation of the origins, with or without that mishap, and Chen Zhu’s feedback would have already been less than enthusiastic without that extra complication. **But, from that time on, the Tedros proposal was basically dead, and the coming WHO mission was limping even more.**

Panic and decision to go for publication to support the WHO (8-10 Feb)

On 8 Feb, Tedros made some rather out of line comments for a normally temperate character:

At WHO, we’re not just battling the virus; we’re also battling the trolls and conspiracy theorists that push misinformation and undermine the outbreak response. [..]

On 8–10 Feb, Farrar decided to turn the confidential report that was meant to be discussed with an ad hoc WHO group, into a published article. **This confirmed that the Tedros proposal was still-born**, as it would have required instead to keep the report confidential, and for the benefit of the WHO committee that would onboard the origin question.

- **The real reason for that decision** was, as we have seen, the premature death of the Tedros proposal. What was left, was for Tedros and Farrar to urgently appease China so that the WHO may soon be allowed in China and may get some proper access to data once there. This was a question of credibility for the WHO, which until now had looked rather weak in its handling of China.
- **The proximal (i.e. pretext) reason** was offered either unwittingly or opportunistically by Holmes to Farrar, who needed little convincing given the WHO predicament.

ment: Holmes, who was busy working on pangolin papers at the time, saw the announcement on 7 Feb of a pangolin virus with a (false) 99% homology, as a good enough pretext to conclude in favour of a zoonotic event.

In short, on 8 Feb, Farrar ordered that the P.O. report should be published (if possible in time for the coming WHO mission), defined the product ('go to scientific statement to refer to') and expressed his desire for a tone ('definitive'). He also shared the draft of P.O. with the WHO, then came back with some requests for changes.

There had never been much chance of any origin investigation during the coming WHO mission, and there was even less now. On 9 Feb, in an interview directed at China, Schwartländer (Tedros' right arm) discussed the coming mission as having probably no access to Wuhan, no real need for it, and with no mention at all of any origin investigation.

LeDuc asks pointed questions and death of the Track II initiatives (9-10 Feb)

At the same time, another event would again have had alert lights flashing in China. On 9 Feb, so just three days after Fauci's inopportune quote in a misleading ABC article, James LeDuc — who was working on strengthening Chinese biosafety standards as part of a Track II effort — sent some pointed questions to Yuan Zhiming, director of the WIV BSL-4 lab, about his laboratory biosafety record. He also encouraged him to fess up to any shortcoming:

I just think that we need to aggressively address these rumours and presumably false accusations quickly and provide definitive, honest information to counter misinformation. If there are weaknesses in your program, now is the time to admit it and get them corrected.

The very next day (10 Feb), the Track II initiative that the US had been trying to push to get access to isolates, officially hit the wall. Chunli Bai, the President of the Chinese Academy of Sciences (CAS), emailed Hamburg and the other NASEM members who had worked in trying to get isolates from China. He informed them that the National Health Commission — a political body directly under the State Council — would take the lead on that sharing.

In his email, Chunli Bai also made the point that the sharing of isolates would have to happen via the WHO, under a WHO framework. In a rather humiliating sentence, Chunli Bai added that many other countries, not just the US, had also asked for isolates.

Chunli Bai to Hamburg: The National Health Commission of China (NHC), [...] will take the lead in coordinating discussions and collaborations with the World Health Organization (WHO) for the sharing of 2019nCov isolates under WHO framework. As you may understand, in addition to the U.S., many other countries have also made requests on isolates sharing from China.

So by now the Tedros proposal was dead, the Track II path was closed, China was redirecting all efforts to the WHO, and the WHO was literally scrambling to get access. China held all the cards.

Renewed need to support the WHO mission (16–17 Feb)

On 16 Feb, Farrar needed another demonstration of support to China to smooth out the (non-investigative) work of the WHO mission, that had just arrived in Beijing that day.

Accordingly:

- On 16 Feb, Farrar worked to upgrade Daszak's Statement of Support from an initially planned release on change.org, to a publication in the top medical journal, *Lancet*.
- At the same time, Farrar made sure that what was now known as Proximal Origin would be released quickly on virological.org on 17 Feb, while it was sent to *Nature* the same day, for peer review, and hopefully a quick publication.

To achieve this doublet of pro-zoonosis publications, Farrar used his personal contacts with both publishers (*Nature* and *Lancet*).

These initiatives were made only more relevant as some non-natural origin questions hit again the headlines in the US by the end of the day (16 Feb US), following yet another overblown controversy around Senator Tom Cotton, who was essentially asking the same questions as the Farrar core-team, and as Baric during his BioSecurity meeting three weeks ago, with the only difference being that Senator Cotton was doing so publicly.

Release of Proximal Origin on virological.org (17 Feb)

On 17 Feb, Rambaut released the draft of P.O. on virological.org. That draft still had a bit of wiggle room for the possibility of passaging, but in fact Andersen and Rambaut had chosen by then not to reflect the true extent of their concerns about that scenario. An extent that they were still demonstrating in their private Slack discussions, and that they would keep displaying privately for months.

Nevertheless, Farrar still asked Andersen to change some wording on that 17 Feb, which further watered down the possibility of a lab construct (from *unlikely* to *improbable*), possibly after sharing it with the WHO.

Passaging and a torpedoed review at *Nature* (20 Feb)

When Farrar had decided to go for publication (8–10 Feb), Drosten, Fouchier and Koopmans had opposed that decision. Their shared concern was that the possibility of passaging, even if very much watered down by then, was still being discussed; something which in their mind was simply unacceptable, as it could re-open a very painful public debate about gain-of-function research, that had seemed closed since 2017.

Against expectations, on 20 Feb *Nature* answered that it would not publish the Proximal Origin manuscript. An anonymous email to Jon Cohen (a writer for *Science* magazine) would later attribute P.O.'s rejection to its lack of acknowledgement of the essential role of

Fouchier and Drosten in vehemently pushing back against the initial acceptance of a possible lab origin, thus not giving proper credit to those who, in the background, ended up shaping P.O. conclusions away from the initial position of Farrar's tight group of co-authors.

However, the review notes (released much later) show that one reviewer was quite positive, while another was taking the rather extreme Fouchier / Drosten / Koopmans line, by wanting P.O. to categorically close any possibility of a lab origin, arguing wrongly that the pangolin data that should soon be released would close the case. Nature seems to have wisely decided that, given the very different reviews and the expectations that one reviewer had about future data, it would rather pass.

Publication in Nature Medicine (17 Mar)

Eventually, to secure publication in Nature Medicine, a lower rank journal in the Nature group, the P.O. authors had to remove any possibility of passaging, which they still considered it a real possibility in their private conversations.

So far, the P.O. co-authors, especially Andersen, Garry and Rambaut, had accommodated the WHO and NIH concerns relayed by Farrar with great dedication. They had taken logical shortcuts, they had banked aggressively on some pangolin sequences they had no good reason to trust, they had tuned down their language when asked, but they still had managed to leave a kernel of scientific truth by keeping the possibility of passaging slightly open, for those who read carefully. That now had to go.

The change, which would have satisfied Koopmans and Fouchier, made it possible to get P.O. published on 17 Mar, albeit with a full-month delay.

With the publication of P.O. in Nature Medicine,

- Andersen and Rambaut had lost control of the process and had folded, faced with the imperatives of Farrar.
- Holmes had reasons to be satisfied with the whole process, being dead-sure of his pangolin hypothesis and with yet another paper with his name on.
- Lipkin had some reservations, but kept them quiet for the moment.

Unexpected success and lingering doubts

In the months following publication in Nature Medicine, Andersen, Holmes, Rambaut and Garry appeared to greatly enjoy their accidental glory, as Proximal Origin — for which they claimed all credit s— became one of the most influential scientific articles in years.

Nevertheless, in mid-April, one month after publication, Andersen would privately still consider passaging as a possibility and would even show second thoughts about engineering. Andersen's lingering doubts were reinforced by the clear evidence of virus dis-

covery and culturing work done at the wholly unsuitable P2 level, and these doubts came again to the surface when newly disclosed State Department cables demonstrated US concerns about biosafety at the WIV already back in 2018.

Lipkin, a good friend of Baric and the last addition to Farrar's core-team, would later mention his own doubts about the wording and logical rigour of Proximal Origin. When faced with another controversial pro-zoonosis paper, Worobey *et al.* (2022), which Holmes offered him to join as a co-author, he would refuse, in an episode that marked the end of their amicable relationship.

Egomania, funding woes and some old foes (Jul)

Still, that is not the end of the P.O. story. In July 2020, as mentioned earlier, Jon Cohen (at Science) received an anonymous email denouncing the P.O. authors as frauds for not acknowledging the key contributions of Fouchier and Drosten. The anonymous email also attributed the rejection by Nature to that authorship issue.

The email author clearly had access to inside information, as there was no way at the time to know what the dynamic of the Farrar call had been (that information would only be released much later). The email author must also have had some rather extraordinary motivations, that seem to go beyond a pure scientific authorship dispute.

Any interpretation of the email to Jon Cohen is necessarily largely speculative, in particular as to whom may have been involved with its writing. Nevertheless, one may note that Daszak was on very bad terms with three of the co-authors of Proximal Origin, namely Holmes, Andersen and Rambaut.

This acrimony was due to the fact that these three had very publicly denounced Daszak's Global Virome Project (GVP), in an article in Nature in 2018, and had criticised PREDICT and the GVP again as recently as Oct 2019. The GVP was the most ambitious project proposal in Daszak's career with a huge price tag (\$1.3 bln), which Daszak had been pushing since 2017. During these high-profile spats in scientific publications, in newspaper articles and on social media, Holmes, Andersen and Rambaut had assaulted Daszak's projects aimed at extensive sampling in the wild, describing them as a misuse of research funds.

Following the unexpected success of Proximal Origin, Holmes, Andersen and Rambaut were now in an even better position to threaten the GVP project and Daszak's grants. All at the precise time when Daszak's NIH grants were already getting under a lot of pressure.

In Daszak's mind, himself, Fouchier and Drosten were the true heroes of this story, while Holmes, Andersen, Gary and Rambaut were clearly impostors, who not only collected the benefits of the insights of others (Fouchier and Drosten), and had eventually caused the suspension of his own R01 grant, but were now in an even better position to sever his GVP lifeline. With the added irony that Daszak had actually contributed to their unexpec-

ted success when he helped establish the narrative with his ‘Statement of Support’, for which — double irony — he could not even claim credit, having had to hide his essential role.



References

1. [/summary-proximal-origin-a-tragicomedy-of-our-times-1ed5b4dae506 ↗](#)
2. [/full-text-proximal-origin-a-tragicomedy-of-our-times-draft-p1-626c72cb05b1 ↗](#)
3. [/full-text-proximal-origin-a-tragicomedy-of-our-times-draft-p2-39b90684bf01 ↗](#)
4. [https://www.altmetric.com/details/77676422/citations?utm_source=altmetric_explorer_details&utm_medium=referral#score ↗](https://www.altmetric.com/details/77676422/citations?utm_source=altmetric_explorer_details&utm_medium=referral#score)
5. [/full-text-proximal-origin-a-tragicomedy-of-our-times-draft-p1-626c72cb05b1 ↗](#)